

$$\therefore \vec{a} \cdot \vec{a} = |\vec{a}|^2$$

Given that magnitude of $\vec{a}$ and $\vec{b}$ is 4

$$\text{i.e., } |\vec{a}| = |\vec{b}| = 4$$

$$\therefore |\vec{a}|^2 = 4^2 = 16$$

$$\text{and } \vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \frac{\pi}{3}$$

$$= 4 \cdot 4 \cdot \frac{1}{2}$$

$$= 8$$

$$|\vec{a} - \vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2 - \vec{a} \cdot \vec{b} - \vec{b} \cdot \vec{a}$$

$$= 16 + 16 - 8 - 8$$

$$= 16$$

$$\cos \theta = \frac{16 - 8}{4 \cdot \sqrt{16}} = \frac{8}{4 \cdot 4} = \frac{1}{2}$$

$$\theta = \frac{\pi}{3}$$

71. Option (a) is correct.

$$\text{Given, D.E: } \frac{dy}{dx} = x$$

$$\Rightarrow \int dy = \int x dx$$

$$\Rightarrow y = \frac{x^2}{2} + C$$

$$\Rightarrow y_1(x) = \frac{x^2}{2} + C_1 \quad \dots(i)$$

$$\text{and } y_2(x) = \frac{x^2}{2} + C_2 \quad \dots(ii)$$

$$\text{For } y_1(0) = 0 \text{ in equation (i)}$$

$$\Rightarrow 0 = 0 + C_1$$

$$\Rightarrow C_1 = 0$$

$$\Rightarrow y_1(x) = \frac{x^2}{2} \quad \dots(iii)$$

$$\text{For } y_2(0) = 0$$

$$\Rightarrow 4 = 0 + C_2 \quad [\text{from (ii)}]$$

$$\Rightarrow C_2 = 4$$

$$\Rightarrow y_2(x) = \frac{x^2}{2} + 4 \quad \dots(iv)$$

So, for number of points of intersection for $y_1(x)$ and $y_2(x)$ solving (iii) and (iv)

$$\frac{x^2}{2} = \frac{x^2}{2} + 4 \quad (\text{Not possible})$$

$$\text{as } 0 \neq 4$$

$\therefore$ No point of intersection.

72. Option (c) is correct.

$$\text{Given, } y = e^x(a \cos x + b \sin x) \quad \dots(i)$$

$$\frac{dy}{dx} = e^x(-a \sin x + b \cos x) + (a \cos x + b \sin x)e^x$$

$$\frac{dy}{dx} = e^x(-a \sin x + b \cos x) + y \quad [\text{using (i)}] \dots(ii)$$

Again differentiate w.r.t. to x ,

$$\frac{d^2y}{dx^2} = -e^x(a \cos x + b \sin x) + (-a \sin x + b \cos x)e^x + \frac{dy}{dx}$$

$$\frac{d^2y}{dx^2} = -y + \frac{dy}{dx} - y + \frac{dy}{dx} \quad [\text{Using (i) and (ii)}]$$

$$\frac{d^2y}{dx^2} - \frac{2dy}{dx} + 2y = 0 \text{ is the required differential equation.}$$

73. Option (d) is correct.

$$f(x) = ax - b \quad \dots(i)$$

$$\text{and } g(x) = cx + d \quad \dots(ii)$$

$$f[g(x)] = g[f(x)]$$

$$\Rightarrow a[g(x)] - b = c[f(x)] + d$$

$$\Rightarrow a(cx + d) - b = c(ax - b) + d \quad [\text{using (i) and (ii)}]$$

$$\Rightarrow acx + ad - b = acx - bc + d$$

$$\Rightarrow ad - b = d - bc$$

$$\Rightarrow ad - b + bc + d = d + d$$

$$\therefore f(d) + g(b) = 2d \quad [\because (d) = ad - b \text{ and } g(b) = bc + d]$$

74. Option (b) is correct.

$$\text{Let } I = \int_{-1}^1 (3 \sin x - \sin 3x) \cos^2 x dx$$

$$I = \int_{-1}^1 (3 \sin x - 3 \sin x + 4 \sin^3 x) \cos^2 x dx$$

$$[\because \sin 3x = 3 \sin x - 4 \sin^3 x]$$

$$I = \int_{-1}^1 4 \sin^3 x \cos^2 x dx$$

$$= \int_{-1}^1 f(x) dx$$

Here, $f(x) = 4 \sin^3 x \cos^2 x$ is an odd function
as $f(-x) = -f(x)$

$$\Rightarrow I = \int_{-1}^1 4 \sin^2 x \cos^2 x dx = 0$$

$$[\because \int_{-a}^a f(x) dx = 0, \text{ if } f(-x) = -f(x)]$$

75. Option (d) is correct.

Given

$$\left\{ 2 - \left(\frac{dy}{dx} \right)^2 \right\}^{0.6} = \frac{d^2y}{dx^2}$$